

Compact Kähler 3-folds with the action of an abelian group of maximal dynamical rank

Guolei Zhong

Center for Complex Geometry, Institute for Basic Science

Zoom Algebraic Geometry Seminar, 15th March 2022

Contents

1. Dynamical degree & Entropy
2. Background & Previous results
3. Projective setting
4. Main result and proofs
5. Open questions

Dynamical degree & Entropy

- (X, ω) : a compact Kähler manifold of $\dim n$ with a Kähler form ω .
- $\text{Aut}(X)$: the group of all holomorphic automorphisms of X .
- The p -th dynamical degree $d_p(g)$ of $g \in \text{Aut}(X)$ is defined as

$$d_p(g) := \max\{|\lambda| : \lambda \text{ is an eigenvalue of } g^*|_{H^{p,p}(X, \mathbb{C})}\},$$

i.e., the spectral radius of the induced linear map $g^*|_{H^{p,p}(X, \mathbb{C})}$.

$$d_p(g) = \lim_{k \rightarrow \infty} \left(\int_X (g^k)^*(\omega^p) \wedge \omega^{n-p} \right)^{1/k}$$

- $h_a(g) := \max_{0 \leq p \leq n} \log d_p(g)$: the algebraic entropy of g .

Theorem (Gromov and Yomdin)

The algebraic entropy coincides with the topological entropy $h_t(g)$, a dynamical invariant measuring the divergence of the orbits.

Dynamical degree & Entropy

Property – Concavity of dynamical degrees (Khovanskii and Teissier)

$\exists u \leq v$ s.t. $1 = d_0(g) < \cdots < d_u(g) = \cdots = d_v(g) > \cdots > d_n(g) = 1$.

- $g \in \text{Aut}(X)$ is of **positive entropy**, if $h_t(g) > 0$, or equivalently $d_1(g) > 1$. A subgroup $G \subseteq \text{Aut}(X)$ is of positive entropy, if every non-trivial element in G is of positive entropy (Dinh-Sibony, 2004).

Definition (Dynamical degree in the singular setting)

Let X be a projective variety (or compact Kähler space).

- $d_1(g) :=$ the spectral radius of g^* on the Néron-Severi space $\text{NS}_{\mathbb{R}}(X)$ (or Bott-Chern cohomology space $H_{\text{BC}}^{1,1}(X)$).
- g is of **positive entropy** if $d_1(g) > 1$, otherwise it is of **zero entropy**.
- By (Dinh-Sibony, 2005), the dynamical degrees can also be defined for dominant meromorphic maps and they are bi-meromorphic invariants.

Background & Previous results

Motivated by Smale's work, Dinh and Sibony studied the structure for an abelian subgroup of $\text{Aut}(X)$.

Theorem (Dinh-Sibony, 2004)

Let $G \subseteq \text{Aut}(X)$ be an abelian subgroup where X is a compact Kähler manifold of $\dim n \geq 2$. Then

- The subset $U \subseteq G$ of the elements of zero entropy is a subgroup.
- There is a free abelian subgroup $H \subseteq G$ which is of positive entropy and has $\text{rank } H \leq n - 1$ such that $G \cong U \times H$.
- If $\text{rank } H = n - 1$ (being maximal), the group U is finite.

Example: complex torus; (Dinh-Sibony, 2004)

The rank estimates in the above theorem are optimal. As mentioned in (Dinh-Sibony, 2004), for each n , there is an n -dim complex torus T such that $\text{Aut}(T)$ contains $G \cong \mathbb{Z}^{\oplus n-1}$, of positive entropy.

Background & Previous results

Zhang proved a theorem of Tits type for compact Kähler manifolds; meanwhile, Dinh-Sibony's rank estimates were thus extended.

Theorem (Zhang, 2009; cf. Campana-Wang-Zhang, 2014)

Let $G \subseteq \operatorname{Aut}(X)$ be a subgroup where X is a compact Kähler manifold of $\dim n \geq 2$. Then either G contains a free non-abelian subgroup $\mathbb{Z} * \mathbb{Z}$, or there is a finite-index solvable subgroup H such that the subset N of zero entropy elements of H is normal in H and H/N is free abelian of rank $\leq n - 1$. In particular, $\operatorname{Aut}(X)$ satisfies Tits alternative.

- Tits' theorem: every linear group G over $\mathbb{R}$ satisfies Tits alternative, i.e., any subgroup of G either has a free non-abelian subgroup, or is virtually solvable (possesses a finite-index solvable subgroup).
- Tits alternative for $\operatorname{Aut}(X)$ was first conjectured in the joint work (Keum-Oguiso-Zhang, 2009).
- (Hu, 2020) extends Zhang's Tits alternative to $\operatorname{char} > 0$.

Background & Previous results

Main question (Dinh-Sibony, 2004; Dinh-Hu-Zhang, 2015)

Classify n -dim compact Kähler manifolds admitting a free abelian group of automorphisms of rank r ($r \leq n - 1$) which is of positive entropy.

Theorem (surface case; Cantat, 1999)

If X is a smooth complex compact surface (Kähler or not) admitting an automorphism of positive entropy, then X is either a complex torus, an Enriques surface, a K3 surface or a non-minimal rational surface.

Theorem (Projective case with maximal dynamical rank; Zhang, 2016)

Let X be a normal projective variety of $\dim n \geq 3$ admitting the action of the group $G := \mathbb{Z}^{\oplus n-1} \subseteq \text{Aut}(X)$ of positive entropy. If X is not rationally connected, then with G replaced by a finite-index subgroup, X is birational to a quotient T/F of a complex torus T (G -equivariantly).

Background & Previous results

- Zhang also treated the case when G is virtually solvable.
- There is a more complete version of the above theorem in (Zhang, 2016), where some conditions can be further weakened.

Example: rational 3-fold; (Oguiso, 2014), (Zhang, 2016)

There exists a rational 3-fold X with at worst klt singularities such that (i) X is a quasi-étale quotient of a 3-dim abelian variety and (ii) X admits an abelian subgroup $G \cong \mathbb{Z}^{\oplus 2} \subseteq \text{Aut}(X)$, which is of positive entropy and of maximal dynamical rank.

Definition (G -equivariancy)

A rational (resp. meromorphic) map $(X, G) \dashrightarrow Y$ of projective varieties (resp. compact Kähler spaces) is **G -equivariant** if G descends to a biregular action $G|_Y$ on Y .

Background & Previous results

The existence of the G -equivariant resolution has been confirmed.

Equivariant resolution (Hironaka, 1964; Włodarczyk, 2009)

Let X be an analytic space defined over $\mathbb{C}$. There exists a canonical desingularization of X that is a manifold $\tilde{X}$ together with a proper bimeromorphic holomorphic map $\text{res}_X : \tilde{X} \rightarrow X$ such that

- $\text{res}_X : \tilde{X} \rightarrow X$ is an isomorphism over the nonsingular locus of X .
- The inverse image of the singular locus $\text{res}_X^{-1}(X_{\text{Sing}})$ is a simple normal crossing divisor.
- res_X is functorial with respect to local analytic isomorphisms. For any local analytic isomorphism $\phi : X' \rightarrow X$ there is a natural lifting $\tilde{\phi} : \tilde{X}' \rightarrow \tilde{X}$ which is a local analytic isomorphism.

Remark: The existence of resolutions of singularities of analytic spaces (in $\text{char} = 0$) was confirmed by Hironaka. The proof has been simplified by Bierstone-Milman and Villamayor in their series of works.

Projective setting (Zhang's program)

G -equivariant fibration (Zhang, 2009; Hu, 2020)

Let (X, G) be a pair where $G \subseteq \operatorname{Aut}(X)$ is a free abelian group of positive entropy. If there is a G -equivariant surjective morphism $\pi : X \rightarrow Y$ of projective varieties (or compact Kähler spaces) with $\dim X > \dim Y > 0$, then $\operatorname{dr}(G|_X) \leq \operatorname{dr}(G|_Y) + \dim X - \dim Y - 1$ ($\operatorname{dr}(G|_Y)$ is the rank of positive entropy part).

A brief review of Zhang's program:

- If X is uniruled, $G|_X$ descends to biregular $G|_Y$ via Nakayama's special MRC fibration $X \dashrightarrow Y$. Applying the above result to a G -equivariant resolution, one can show X is rationally connected.
- If X is not uniruled, then K_X is pseudo-effective (BDPP, 2013), and one can run the G -equivariant minimal model program with scaling.

Remark: Barlet's analytic cycle theory $\Rightarrow$ R.C. of uniruled Kähler 3-folds (with a free abelian group of maximal rank); cf. (Brunella, 2006).

Projective setting (How to make sure the G -equivariancy?)

One needs an important nef and big class to ensure the G -equivariancy.

Theorem (Dinh-Sibony, 2004; Zhang, 2009)

Let X be a normal projective variety (or a compact Kähler manifold) of $\dim n$ and $G = \mathbb{Z}^{\oplus r} \subseteq \text{Aut}(X)$ a free abelian group of positive entropy ($r \leq n - 1$). Then there are nef $\mathbb{R}$ -Cartier divisors (resp. nef $(1, 1)$ -classes) $L_1, \dots, L_{r+1}$ such that

- each L_i is a common eigenvector of G , i.e., there are $r + 1$ characters $\chi_i : G \rightarrow \mathbb{R}_{>0}$ such that $g^* L_i = \chi_i(g) L_i$ for any $g \in G$; and
 - $L_1 \cdots L_{r+1} \neq 0$ as an element in $H^{r+1, r+1}(X, \mathbb{R})$.
-
- In particular, if $r = n - 1$ (being maximal), then we get a nef and big $\mathbb{R}$ -Cartier $\mathbb{R}$ -divisor (resp. $(1, 1)$ -class) $A := \sum_{i=1}^n L_i$.
 - (Property of A ; Zhang, 2016) For every G -periodic (k, k) -class $\alpha \in H^{k, k}(X, \mathbb{R})$, $A^{n-k} \cdot \alpha = 0$.

Projective setting

Using such A , we continue to recall Zhang's program for the pure case.

Proof of the G -equivariancy (Zhang, 2016)

- Finiteness of $(K_X + A)$ -neg. ext. rays $\Rightarrow$ Assume each $(K_X + A)$ -neg. ext. ray is A -trivial $\Rightarrow$ the G -image of a $(K_X + A)$ -neg. ext. ray is still $(K_X + A)$ -negative $\Rightarrow$ the G -equivariancy $\Rightarrow A$ is the pull-back of a nef $\mathbb{R}$ -Cartier $\mathbb{R}$ -divisor from downstairs by the cone theorem.
- (BCHM, 2010) $\Rightarrow$ such G -MMP with scaling will terminate in finitely many steps $\Rightarrow$ the end product Y with $K_Y + A_Y$ being nef $\Rightarrow$ may assume $K_Y + A_Y$ is nef and big.
- Base-point-free theorem (BCHM, 2010) $\Rightarrow$ a birational morphism $Y \rightarrow Z$ (also G -equivariant) with $K_Z + A_Z$ being ample.
- σ -decomposition for $K_Z \Rightarrow K_Z \sim_{\mathbb{Q}} 0$ and A_Z is thus ample.
- $A_Z^{n-2} \cdot c_2(Z) = 0 \Rightarrow c_2(Z) = 0$ (Miyaoka, 1985) $\Rightarrow Z$ is a Q -abelian variety (Yau, 78; Greb-Kebekus-Peternell, 16; Lu-Taji, 18).

Main result and proofs

Hypothesis: Let (X, D) be a $\mathbb{Q}$ -factorial compact Kähler klt 3-fold pair. Suppose there is an action of a free abelian group $G := \mathbb{Z}^2 \subseteq \text{Aut}(X)$, of positive entropy and of maximal rank. Suppose further every irreducible component of the support of D is G -periodic.

Theorem (Z., 2022)

Let (X, D, G) be a pair satisfying Hypothesis. If X is not rationally connected, then with G replaced by a finite-index subgroup, we have

- 1 There is a G -equivariant bimeromorphic map $X \dashrightarrow Z$ such that Z is a quasi-étale quotient of a complex 3-torus.
- 2 $Z \cong T/H$ where H is a finite group acting freely outside a finite set of a complex 3-torus T . Moreover, the quotient morphism $T \rightarrow Z$ is also G -equivariant.
- 3 There is no positive dim G -periodic proper subvariety of Z ; in particular, the bimeromorphic map $X \dashrightarrow Z$ in (1) is holomorphic.

Main result and proofs

Rationally connected compact Kähler manifolds being projective, we immediately have the following corollary.

Corollary

If X is a non-algebraic compact Kähler 3-fold admitting an action of a free abelian group $G := \mathbb{Z}^2 \subseteq \text{Aut}(X)$ of positive entropy, then X is bimeromorphic to a quotient of a complex 3-torus, with the quotient morphism being (G -equivariantly) étale outside finitely many points.

Applying (Dinh-Sibony, 2004) to $\pi^* \text{Nef}(X)$, the pull-back of the nef cone along a G -equivariant resolution $\tilde{X} \rightarrow X$, we get three G -common nef eigenvectors ξ_1, ξ_2, ξ_3 with the sum $\xi := \xi_1 + \xi_2 + \xi_3$ a nef and big class.

Question

Are there only finitely many $(K_X + \xi)$ -negative (or non-positive) extremal rays in $\overline{\text{NA}}(X)$? Are they generated by rational curves?

A brief review on the MMP for Kähler varieties ($\dim \leq 3$)

- (Fujino, 2021) Fujino established the MMP for $\mathbb{Q}$ -factorial log surfaces in Fujiki's class.
- (Filip-Tosatti, 2018) proved the base point free theorem in $\dim 2$.
- (Höring-Peternell, 2015) & (Höring-Peternell, 2016) established the MMP for $\mathbb{Q}$ -factorial terminal Kähler 3-folds.
- (Campana-Höring-Peternell, 2016; 2021) proved the abundance for compact Kähler terminal 3-folds except that X is a simple 3-fold of Kodaira dim zero.
- (Das-Ou, 2022) proved the log abundance for compact Kähler log canonical 3-fold pair (X, Δ) with $K_X + \Delta$ being nef and the num dim $\nu(K_X + \Delta) \neq 2$.
- (Campana-Höring-Peternell, 2016), (Das-Ou, 2022) and (Das-Hacon, 2022) confirmed existence of the MMP for dlt pairs on Kähler 3-folds.
- (Das-Hacon, 2022) proved the base point free theorem on Kähler 3-folds; cf. (Höring, 2018) and (Tosatti-Zhang, 2018).

Main result and proofs

Proposition (finiteness of $(K_X + \xi)$ -negative extremal rays; Z., 2022)

Let X be a normal $\mathbb{Q}$ -factorial compact Kähler 3-fold and Δ an effective $\mathbb{Q}$ -divisor such that (X, Δ) has only klt singularities. Let α be a nef and big class on X . Suppose that one of the following conditions holds.

- ① $K_X + \Delta$ is pseudo-effective; or
- ② $K_X + \Delta$ is not pseudo-effective but $K_X + \Delta + \alpha$ is a big class.

Then there are only finitely many $(K_X + \Delta + \alpha)$ -negative extremal rays, all of which are generated by rational curves.

- Our proof relies on the cone theorem. Under the above condition, there is a fixed number d , such that every $(K_X + \Delta)$ -negative extremal ray R is generated by a curve ℓ with $-(K_X + \Delta) \cdot \ell \leq d$.
- The cone theorem (✓) for the non-uniruled 3-fold case, but there are some differences with the projective case when X is uniruled.

Main result and proofs

Proposition (End product of the G -equivariant $(K_X + \xi)$ -MMP)

If $K_X + \xi$ is nef, there is a G -equivariant bimeromorphic map $\tau : X \rightarrow Y$ such that Y is a quotient of a torus.

- With ξ replaced by a multiple $\Rightarrow K_X + \xi$ is nef and big and every $(K_X + \xi)$ -trivial ray R is also ξ -trivial.
- Base point free theorem (Das-Hacon, 2022) $\Rightarrow$ there is a holomorphic map $\tau : X \rightarrow Y$ with connected fibre such that $K_X + \xi = \tau^*\beta$ for some Kähler class β on Y .
- τ is G -equivariant, but the proof is a bit more troublesome when compared with the projective case. We skip the technical proof here.
- τ is ξ -trivial $\Rightarrow \xi = \tau^*\xi_Y \Rightarrow K_X = \tau^*K_Y$ with $\beta = K_Y + \xi_Y$ Kähler.
- (Lemma) $\text{Null}(\xi_Y)$ (union of the positive dim proper analytic subvariety V such that $\xi_Y|_V$ is not big) = union of the G -periodic proper analytic subvariety.

Main result and proofs

- Let $\pi_1 : Y_1 \rightarrow Y$ be a G -equivariant resolution with the non-Kähler loci $E_{\text{nK}}(\pi_1^* \xi_Y) = \text{Null}(\pi_1^* \xi_Y)$ (Collins and Tosatti, 2015).
- By (Demailly, 1992) and (Boucksom, 2004), we may choose a Kähler current T in $\pi_1^* \xi_Y$ with analytic singularities along $E_{\text{nK}}(\pi_1^* \xi_Y)$.
- Blowing-up the ideal sheaf and taking Hironaka's resolution, we have $\pi_2 : Y_2 \rightarrow Y_1$ such that $\pi^* \xi_Y = \omega + E$ (with $\pi := \pi_1 \circ \pi_2$) where ω is a Kähler class and $E \geq 0$ is an $\mathbb{R}$ -divisor, $\text{Supp}(\pi_2(E)) = E_{\text{nK}}(\pi_1^* \xi_Y)$.
- Hodge index theorem $\Rightarrow \xi_Y|_V \equiv 0$ for every positive dim G -periodic proper subvariety V .
- $\pi^* \xi_Y|_E \equiv 0 \Rightarrow \pi^* K_Y \cdot \omega^2 = 0$ (Hodge-Riemann) $\Rightarrow K_Y \equiv 0$, noting that K_Y is psuedo-effective $\Rightarrow \xi_Y$ is a Kähler class.
- $c_2(Y) \cdot \xi_Y = 0$; $c_2(Y)$ the orbifold second Chern class (Graf-Kirschner, 2020) $\Rightarrow Y$ is a Q -complex 3-torus by (Graf-Kirschner, 2020) (in the Kähler setting). The proposition is proved. (✓)

Main result and proofs

Some remarks:

- The G -equivariancy of the quotient morphism $T \rightarrow Y$ shares almost the same proof as in the projective setting, so we skip it.
- Graf and Kirschner's characterization of Q -complex 3-tori has been recently extended to arbitrary dim, assuming the smoothness in codim 2; cf. (Claudon-Graf-Guenancia, 2021).
- (Claudon-Graf-Guenancia, 2021) also showed the semi-positivity of c_2 for a general compact Kähler space X with $c_1 = 0$, which is klt and smooth in codimension two.

Summarize the proofs

- (1). Start from the initial X and run $(K_X + \xi)$ -MMP G -equivariantly.
- (2). End up with some X_n such that $K_{X_n} + \xi_{X_n}$ is nef.
- (3). Such X_n admits a G -equivariant bimeromorphic holomorphic map to a quotient Y of a complex torus T .

$$X \dashrightarrow \cdots \dashrightarrow X_n \rightarrow Y \cong T/F.$$

Open question

In the view of Dinh-Sibony and Zhang's works, we would like to know the general case when G is assumed to be a free abelian group with lower rank.

Question 1

Let X be either a compact Kähler manifold or a projective variety of dimension n . Suppose X admits the action of a free abelian group $G \subseteq \operatorname{Aut}(X)$, of positive entropy and of rank $r \leq n - 1$. Can we classify such X ?

(Hu-Li, 2020) partially answered the above question when $\operatorname{rank} G = n - 2$.

In the view of Oguiso's example on the rational klt 3-fold being a quotient of an abelian variety, we ask the following question.

Question 2; cf. (Zhang, 2016)

Does there exist a pair (X, G) with G being of maximal rank such that X is rationally connected but X is not bimeromorphic to a quotient of a complex torus?

Thank you for your attention!